

Laboratory Tests and Faecal Markers in Paediatric Gastroenterology.

A thirteen-year-old boy has had three months of loose stools and tummy pain. He has lost a little weight, but he looks well. His family doctor has already sent blood tests, and the letter says, with some relief, that the blood count, the inflammatory markers and the albumin are all normal. Does that mean he does not have inflammatory bowel disease? In the same clinic, an eighteen-month-old girl who is thriving and has slightly loose stools comes with a faecal calprotectin of one hundred and forty micrograms per gram, flagged in red as abnormal. Does she need a colonoscopy? And a third child, with watery diarrhoea, has a faecal elastase of one hundred and twenty. Is her pancreas failing?

This chapter is about the laboratory tests that a paediatric gastroenterologist orders every day: the blood count and inflammatory markers, faecal calprotectin, faecal elastase and a handful of narrower tests. The main idea is simple to say. Every marker measures one biological process, and it is only as good as the question it is asked. A normal blood test does not exclude mild inflammatory bowel disease. Calprotectin is the best test we have to rule inflammatory bowel disease out, but the age of the child, the drugs the child takes and the laboratory kit all move the number. And faecal elastase is a pancreatic test measured in a gut sample, so watery stool and diseases of the small bowel can make it look low when the pancreas is fine.

A short note on the sources. For the diagnosis of inflammatory bowel disease in children, the main document is the revised Porto criteria of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, published by Levine and colleagues in 2014. For management we use the joint guidelines of the European Crohn's and Colitis Organisation, called ECCO, and ESPGHAN: the paediatric ulcerative colitis guideline of 2018 by Turner and colleagues, and the paediatric Crohn's disease guideline of 2021 by van Rheenen and colleagues. For treatment targets we use the international consensus called STRIDE two, published in 2021. In the United Kingdom, the National Institute for Health and Care Excellence, called NICE, published diagnostics guidance on calprotectin in 2013. For adults, the American Gastroenterological Association, called AGA, published two guidelines on biomarkers in 2023, one for Crohn's disease and one for ulcerative colitis. The rest of the numbers come from individual studies, and we will name them as we go.

A few words first. Inflammatory bowel disease, which we will call IBD, includes Crohn's disease and ulcerative colitis. The erythrocyte sedimentation rate, called ESR, measures how fast red cells settle in a tube, which depends mostly on the proteins in the plasma. C-reactive protein, called CRP, is a protein made by the liver in response to inflammation. Faecal calprotectin is a protein released by neutrophils, a type of white blood cell, into the gut.

Faecal elastase is an enzyme made by the pancreas that passes through the gut into the stool. When we talk about tests, sensitivity is the proportion of children with the disease whom the test picks up. Specificity is the proportion of children without the disease whom the test correctly calls negative. The positive predictive value is the chance that a child with a positive test really has the disease, and the negative predictive value is the chance that a child with a negative test really does not. The last two depend heavily on how common the disease is in the group being tested.

Part one. What each marker actually measures.

Before looking at numbers, it helps to know where each marker comes from, because that explains when it fails.

C-reactive protein is made in the liver, and its production is switched on mainly by a signalling molecule called interleukin six. The level rises within about six hours of an inflammatory stimulus and peaks at about forty-eight hours. Its half-life in the blood is about nineteen hours, and that half-life is the same in health and disease. So the only thing that sets the CRP level is how fast the liver is making it. That makes CRP a fast marker. It rises quickly and falls quickly. Its weakness is that some kinds of gut inflammation produce little of the signal that drives it, as we will see in ulcerative colitis.

The ESR depends mainly on plasma proteins such as fibrinogen and immunoglobulins. It rises and falls slowly, and many things move it, including anaemia. Albumin is the balance between what the liver makes and what the body loses. It falls with chronic inflammation and with protein loss through the gut, so it is a late and non-specific marker.

Calprotectin is a calcium-binding protein made of two smaller proteins called S100A8 and S100A9. It makes up about sixty percent of the protein in the cytoplasm of a neutrophil. When neutrophils move into the gut wall and the lumen, as they do in active inflammation, calprotectin ends up in the stool. It is very stable there, for up to seven days at room temperature, and it is spread evenly through the stool. That is why it works so well as a stool test. But anything that brings neutrophils into the gut will raise it, including infection and drugs.

Faecal elastase is made by the acinar cells of the pancreas. It is not broken down as it passes through the gut, so the amount in stool reflects how much the pancreas secretes. But it is reported as a concentration, in micrograms per gram of stool. So if the stool contains a lot of water, the same amount of elastase is spread through more stool, and the result falls.

Finally, alpha one antitrypsin is a plasma protein that resists digestion. If it appears in stool in large amounts, plasma protein must be leaking into the gut, which is the definition of protein-losing enteropathy.

Part two. Mild disease hides from the blood.

Let us start with the boy whose blood tests were normal. The key study was published in Pediatrics in 2007 by Mack and colleagues. They looked at five hundred and twenty-six children with newly diagnosed IBD, three hundred and ninety-two with Crohn's disease and one hundred and thirty-four with ulcerative colitis. They checked four simple tests: haemoglobin, platelet count, ESR and albumin.

If you imagine a bar chart of the proportion of children in whom all four tests were normal, the results are striking. In mild ulcerative colitis, fifty-four percent had completely normal results, more than half. In mild Crohn's disease, twenty-one percent did, about one in five. Only in moderate or severe disease were all four tests normal in very few children, about four percent. The authors concluded that normal laboratory values cannot be relied on to exclude mild IBD.

The point is that normal results are most common exactly where the clinical picture is least obvious. The child with severe disease looks sick and has abnormal blood tests. The child with mild disease may look well and have normal blood tests. And that is the child where we most need help from the laboratory.

Part three. CRP is often silent in colitis.

The second surprise is how often CRP is normal in children with IBD, especially ulcerative colitis. In a Finnish study published in 2010 by Sidoroff and colleagues, thirty-nine children with IBD had their CRP measured at the time of colonoscopy. Sixty-four percent had an undetectable standard CRP, below five milligrams per litre. In Crohn's disease the figure was forty-five percent. In ulcerative colitis it was eighty-four percent. A high-sensitivity CRP assay did not help to separate active from quiet disease.

Why should this be? Remember that CRP is driven by interleukin six. Crohn's disease involves the whole thickness of the bowel wall, and it is thought to drive much more of this signal than ulcerative colitis, which is limited to the mucosa, the inner lining. So a child with bloody diarrhoea and a normal CRP is not reassuring. A normal CRP is the expected result in ulcerative colitis.

Part four. Cheap blood tests, honest limits.

Can a combination of simple blood tests do better? Two studies give a fair answer. Sabery and Bass, writing in Pediatrics in 2007, studied two hundred and twenty-seven children referred with gastrointestinal symptoms, forty of whom had IBD. The combination of a raised ESR and anaemia had a sensitivity of eighty-three percent and a specificity of ninety-six percent. A commercial panel of antibodies, often marketed as a serological test for IBD, had a sensitivity of only sixty percent in the same children. So the cheap tests did better than the expensive ones.

Cabrera-Abreu and colleagues, in the Archives of Disease in Childhood in 2004, looked at one hundred and fifty-three children, one hundred and three of them with IBD. The best rule was that at least one of the haemoglobin and the platelet count was abnormal. That rule had a sensitivity of about ninety-one percent. Even so, the authors said the tests were not completely reliable.

So blood tests raise the probability of IBD when they are abnormal, but even the best combination misses about one child in six. They cannot lower the probability far enough to skip a stool test.

Part five. Calprotectin: the best rule-out test.

Faecal calprotectin is where the evidence is strongest. Several meta-analyses agree. In 2010, van Rheenen and colleagues published a meta-analysis in the British Medical Journal. In three hundred and seventy-one children and teenagers, calprotectin had a sensitivity of ninety-two percent and a specificity of seventy-six percent. They calculated that if calprotectin were used to decide who goes on to endoscopy, the number of endoscopies in children would fall by about sixty-five percent, at the price of a few children being diagnosed later. A meta-analysis by Henderson and colleagues in 2014 found a sensitivity of ninety-eight percent and a specificity of sixty-eight percent. An individual patient data meta-analysis by Degraeuwe and colleagues in 2015, with nine studies and eight hundred and fifty-three children, found a sensitivity of ninety-seven percent and a specificity of seventy percent. At a cut-off of fifty micrograms per gram, seventeen percent of children had a false positive result and only two percent a false negative.

The setting matters a great deal. In a Dutch study of children referred to a specialist, Holtman and colleagues found a sensitivity of ninety-nine percent. In a British primary care study by Walker and colleagues in 2020, with one hundred and ninety-five children aged four to eighteen and an IBD prevalence of only seven percent, a cut-off of one hundred micrograms per gram gave a sensitivity of one hundred percent and a specificity of ninety-one percent. But the positive predictive value was only forty-three percent. In other words, in primary care, more than half of the children with a positive test did not have IBD. The negative predictive value was one hundred percent.

So the pattern is consistent. Calprotectin sensitivity is above ninety percent in nearly every study. Its specificity is seventy to ninety percent. A negative result almost always means no IBD. A positive result means that something is inflamed, and in a low-risk population it is usually not IBD.

Part six. Calprotectin sorts the waiting room.

The most useful way to think about calprotectin is as a sorting tool. Holtman and colleagues published an individual patient data meta-analysis in JAMA Pediatrics in 2017, with eight

studies and one thousand one hundred and twenty children. They asked how well children could be sorted into low, intermediate and high risk of IBD, first using symptoms alone and then adding calprotectin.

Picture a bar chart with two bars for each of three outcomes. With symptoms alone, only thirty-three percent of the children without IBD were correctly placed in the low-risk group. With calprotectin added, ninety-one percent were. With symptoms alone, sixteen percent of children who did have IBD were wrongly placed in the low-risk group. With calprotectin, nine percent were. And most striking of all, with symptoms alone, fifty-five percent of children ended up in the intermediate group, where nobody knows what to do. With calprotectin, only six percent did. Adding calprotectin raised the area under the curve by zero point two six. The next best test, the ESR, added only zero point one six.

The intermediate group is the group that generates repeat visits, repeated blood tests and anxious families. So the real value of calprotectin is not that it makes a diagnosis. It is that it leaves far fewer children in limbo.

A Dutch trial called CACATU, published in Pediatrics in 2020 by Van de Vijver and colleagues, turned this into a strategy. It included one hundred and ninety-three children aged six to eighteen with non-bloody diarrhoea, twenty-two of whom had IBD. Children were referred for endoscopy if their symptoms were combined with a CRP above ten milligrams per litre, a low haemoglobin, or a calprotectin of at least two hundred and fifty micrograms per gram. With this strategy, fourteen of every hundred children went to endoscopy, only three of them without IBD, and no child with IBD was missed.

Part seven. Below four, the cut-off moves.

Now to the eighteen-month-old girl. The usual cut-off of fifty micrograms per gram comes from adults and older children. Healthy infants have much higher values, probably because the gut barrier and the immune system of the infant gut are still maturing.

A meta-analysis published in 2025 pooled twenty-three studies of two thousand eight hundred and eighty-three healthy children aged up to four years. Imagine a bar chart with one bar for each age group, and a horizontal line at fifty. In infants under one month, the mean calprotectin was two hundred and fifty-eight micrograms per gram. From one to six months it was two hundred and thirty-nine. From seven to twelve months it was one hundred and sixteen. In the second year of life it was one hundred and five, in the third year seventy-five, and only in the fourth year, between thirty-seven and forty-eight months, did the mean fall to thirty-four, below the adult cut-off. So healthy young infants have average values about five times the adult cut-off.

Other cohorts agree. In a French study of sixty-nine term newborns, the median calprotectin on day four of life was one hundred and sixty-seven, and only three babies were below fifty.

In a Nordic population cohort, the upper reference limit was six hundred and fifteen at six months of age and fell to fifty-seven at twenty-four months. And even in older children the cut-off of fifty is not perfect. In a Spanish study of two hundred and twelve healthy children aged four to sixteen, the ninety-fifth percentile fell from one hundred and eighty-seven micrograms per gram at age four to sixty-one at age sixteen, and about one in five healthy children had a value above fifty. The authors proposed an age-based nomogram.

So in a child under four, a calprotectin of one hundred and forty is within the range of healthy children. On its own, it is not a reason for colonoscopy. If there are other worrying features, such as poor growth, blood in the stool or low albumin, that is a different matter. But the number alone should not drive the decision.

Part eight. Why calprotectin is high without IBD.

There are four common reasons for a raised calprotectin in a child without IBD.

The first is drugs. Non-steroidal anti-inflammatory drugs such as ibuprofen injure the small bowel. In a study by Tibble and colleagues in *Gut* in 1999, forty-four percent of three hundred and twelve people taking these drugs had a raised calprotectin, and about a fifth of those had levels in the range typical of IBD. Proton pump inhibitors also matter. In a Swedish cross-sectional study of five hundred and ninety adults with a normal colonoscopy, published in 2019, thirty-six percent had a calprotectin above fifty, and taking a proton pump inhibitor nearly quadrupled the odds, with an adjusted odds ratio of about three point eight.

The second is infection. Bacterial gastroenteritis brings neutrophils into the gut and raises calprotectin, so stool pathogens should be checked first, and the test repeated after recovery.

The third is sampling. Calprotectin varies from day to day. In a study of quiet Crohn's disease, the agreement between three samples taken on consecutive days was good, with an intraclass correlation of zero point eight four, but variation was greatest at high values.

The fourth is the assay itself. Different laboratory kits agree well below fifty and above one hundred and fifty micrograms per gram. In the grey zone between fifty and one hundred and fifty, agreement between two kits in a 2022 comparison was poor, with a kappa of only zero point two two. So results from different laboratories cannot be compared directly, and each laboratory's own cut-offs should be used.

The practical rule follows. Many pathways stop non-steroidal drugs for several weeks and, where possible, the proton pump inhibitor before a repeat test, and ask for the first stool of the morning. A result in the grey zone is a reason to remove the confounder and repeat the test, not a reason to scope.

Part nine. From diagnosis to treatment target.

Calprotectin is also a monitoring tool once IBD is diagnosed. The STRIDE two consensus of 2021 sets intermediate treatment targets: a CRP back to normal and a calprotectin reduced to an acceptable range of one hundred to two hundred and fifty micrograms per gram. In children, STRIDE two adds a long-term target of restoring normal growth. The ECCO and ESPGHAN paediatric ulcerative colitis guideline aims for clinical remission, meaning a Paediatric Ulcerative Colitis Activity Index below ten, together with biochemical remission measured by calprotectin. The paediatric Crohn's disease guideline says that serial calprotectin or small-bowel imaging are more reliable markers of response than symptom scores alone.

Why measure, rather than ask how the child feels? Because symptoms and mucosa disagree. In a study of twenty-eight children with ulcerative colitis in clinical remission, published in 2021, only forty-three percent had completely healed mucosa, and half had moderate or severe endoscopic activity.

The adult guidelines go further. The 2023 AGA guideline for Crohn's disease says that in a patient in symptomatic remission, a calprotectin below one hundred and fifty with a normal CRP can rule out active inflammation and avoid an endoscopy. After surgery, a calprotectin below fifty can exclude recurrence in low-risk patients. The ulcerative colitis guideline uses the same cut-off of one hundred and fifty.

The practical message is that a child who feels well can still have active mucosa. Measure calprotectin rather than infer it.

Part ten. Faecal elastase: a pancreas test in a gut sample.

Faecal elastase is the standard screening test for exocrine pancreatic insufficiency. Most laboratories call a value above two hundred micrograms per gram normal, a value between one hundred and two hundred mild to moderate insufficiency, and a value below one hundred severe insufficiency.

In cystic fibrosis it works very well. In an Italian multicentre study by Carroccio and colleagues, published in 2001, with one hundred and thirty-seven subjects, elastase picked up every child with cystic fibrosis and pancreatic maldigestion who was not on enzymes, a sensitivity of one hundred percent. Against healthy children the specificity was also one hundred percent. But against children with diseases of the small intestine, the specificity fell to eighty-six percent. False low results were seen in giardiasis, in short bowel syndrome, in coeliac disease and in intestinal pseudo-obstruction. Low elastase in a child with damaged villi is not proof that the pancreas is failing. Treat the gut disease and recheck.

There are three more practical points. In newborns, elastase is low in meconium, with a mean of about forty-six micrograms per gram, but the second stool already averages about two hundred and forty-three. Values become normal by about day three in term babies and

by about two weeks in very preterm babies born before twenty-eight weeks. So meconium should not be tested. Second, pancreatic enzyme supplements do not interfere with the test. With the polyclonal assay, porcine pancreatin did not cross-react, so enzymes do not need to be stopped. And third, monoclonal and polyclonal kits do not give interchangeable results. In a 2024 comparison the agreement between them was only moderate, with a kappa of zero point four three, and the authors concluded the methods are not commutable.

Part eleven. Water dilutes elastase.

Now to the child with watery diarrhoea and an elastase of one hundred and twenty. Remember that elastase is reported per gram of wet stool. In a German study published in 2001 by Fischer and colleagues, diarrhoeal stool samples were measured as they were and again after freeze-drying, which removes the water. The results after freeze-drying were considerably higher. Eleven percent of the samples that read below two hundred, sixteen of one hundred and fifty-one, were above two hundred once the water was removed. In other words, about one in nine low results in watery stool was simply dilution.

The practical rule is to send a formed stool sample when you can, and to repeat a low result from watery stool before you label a child as having pancreatic insufficiency. A low elastase in a child with secretory diarrhoea is a dilution until proven otherwise.

Part twelve. Narrow tests for narrow questions.

A few other tests are worth knowing, each for one narrow question. Faecal alpha one antitrypsin answers the question of whether protein is being lost into the gut. Its units and cut-offs differ between laboratories, and it needs a fresh sample. Some laboratories measure a twenty-four-hour clearance, which in protein-losing enteropathy is usually above fifty millilitres per day.

Faecal S100A12 is a related neutrophil protein that, in one study by Sidler and colleagues, had a sensitivity and specificity of ninety-seven percent for IBD, but it is not widely available. Faecal lactoferrin is another neutrophil marker, with a wide range of reported accuracy and less stability than calprotectin.

Serological markers can hint at the type of IBD. In a large meta-analysis by Reese and colleagues in 2006, the pattern of positive antibodies to *Saccharomyces cerevisiae*, called ASCA, with negative perinuclear antineutrophil cytoplasmic antibodies, called pANCA, had a sensitivity of about fifty-five percent and a specificity of about ninety-three percent for Crohn's disease. In four hundred and six children with colonic IBD, pANCA was positive in sixty-four percent of those with ulcerative colitis but also in thirty percent of those with Crohn's colitis. The authors concluded that serology cannot reliably tell unclassified IBD from ulcerative colitis or Crohn's colitis. So serology is a footnote to endoscopy and histology, never a substitute.

Part thirteen. What the guidelines say.

On using calprotectin to triage a child with suspected IBD, the documents agree. NICE recommends it in children with suspected IBD who have been referred for specialist assessment, and the adult guidelines endorse it too.

On the cut-off for ruling out disease, they differ. NICE gives no number at all. Paediatric studies use fifty, one hundred or two hundred and fifty, depending on the setting. The adult AGA guidelines use one hundred and fifty, combined with a normal CRP, in patients in symptomatic remission.

On adjusting for age in children under four, there is a gap. None of the guidelines gives age-specific values, even though the physiology is well documented.

On whether normal blood tests exclude IBD, the documents are aligned. None of them accepts normal blood tests as a reason not to investigate.

On endoscopy at diagnosis, there is a material difference. The Porto criteria require ileocolonoscopy and upper endoscopy for every child at diagnosis. The adult guidelines allow biomarkers to guide treatment without endoscopy in some settings. In children, a biomarker can decide who goes to endoscopy, but it does not replace the endoscopy that makes the diagnosis.

On monitoring targets, the documents are aligned around the same idea: clinical remission is not enough, and calprotectin should be part of the target. Finally, none of the IBD guidelines addresses how to interpret faecal elastase, which is a gap.

Part fourteen. Pearls, and what to do differently in clinic.

Let us go back to the three children from the opening.

The thirteen-year-old boy with three months of loose stools and normal blood tests may still have mild IBD. More than half of children with mild ulcerative colitis and a fifth with mild Crohn's disease have entirely normal blood tests at diagnosis. He needs a calprotectin. If it is clearly raised, above about two hundred and fifty, he should have an endoscopy. If it is in the grey zone, check for infection and drugs and repeat it. If it is normal, IBD is very unlikely.

The eighteen-month-old girl with a calprotectin of one hundred and forty who is thriving has a value within the range for healthy children of her age. She does not need a colonoscopy for that number alone.

The child with watery diarrhoea and an elastase of one hundred and twenty may simply have diluted stool. Repeat the test on a formed sample before you start enzymes, and think about the small bowel, because coeliac disease, giardiasis and short bowel can all lower elastase.

So here is what to do differently in clinic. First, never let normal blood tests close the question of IBD. Second, do not be reassured by a normal CRP in a child with colitis. Third, use calprotectin as the main rule-out test, and remember that in a low-risk population a positive result is usually not IBD. Fourth, interpret calprotectin against the child's age, and do not scope a well child under four for a value in the low hundreds. Fifth, before repeating a grey-zone result, stop non-steroidal drugs, consider stopping a proton pump inhibitor, exclude infection and use the same laboratory. Sixth, measure calprotectin during follow-up of IBD, even in children who feel well. And seventh, read faecal elastase against the stool it came from and the state of the small bowel.

Key points.

In children newly diagnosed with IBD, all four basic blood tests were normal in fifty-four percent of those with mild ulcerative colitis and twenty-one percent of those with mild Crohn's disease.

CRP was undetectable in eighty-four percent of children with ulcerative colitis at colonoscopy.

Faecal calprotectin has a sensitivity above ninety percent for paediatric IBD and a specificity of seventy to ninety percent.

Using calprotectin to select children for endoscopy would reduce paediatric endoscopies by about sixty-five percent.

Adding calprotectin to symptoms reduced the proportion of children left at intermediate risk from fifty-five percent to six percent.

Healthy infants have calprotectin values of one hundred to two hundred and fifty micrograms per gram, and values settle only after about three to four years of age.

Non-steroidal drugs, proton pump inhibitors and infection raise calprotectin, and laboratory kits disagree in the zone between fifty and one hundred and fifty.

Faecal elastase is falsely low in watery stool and in meconium, and it can be low in small-bowel diseases such as coeliac disease and giardiasis.

Questions to test yourself.

Question one. In the study by Mack and colleagues, what proportion of children with mild ulcerative colitis had all four basic blood tests normal at diagnosis? Answer: fifty-four percent.

Question two. Why is CRP often normal in ulcerative colitis? Answer: CRP production is driven mainly by interleukin six, and inflammation limited to the mucosa is thought to drive

much less of this signal than the transmural inflammation of Crohn's disease. In one study, CRP was undetectable in eighty-four percent of children with ulcerative colitis.

Question three. What are the pooled sensitivity and specificity of faecal calprotectin for IBD in children in the 2010 meta-analysis by van Rhee and colleagues? Answer: ninety-two percent and seventy-six percent.

Question four. In a primary care study with an IBD prevalence of seven percent, what was the positive predictive value of calprotectin at a cut-off of one hundred? Answer: forty-three percent, so most children with a positive result did not have IBD.

Question five. What is the mean faecal calprotectin in healthy infants aged one to six months, and what does that mean for the cut-off of fifty? Answer: about two hundred and forty micrograms per gram, nearly five times the adult cut-off, so the adult cut-off cannot be used in young children.

Question six. Name two drugs that raise faecal calprotectin. Answer: non-steroidal anti-inflammatory drugs and proton pump inhibitors.

Question seven. Why can faecal elastase be falsely low in watery stool, and how often did this happen in the study by Fischer and colleagues? Answer: elastase is reported per gram of wet stool, so extra water dilutes it. About eleven percent of low results in diarrhoeal stool were normal after freeze-drying.

Question eight. What intermediate target for calprotectin does the STRIDE two consensus set? Answer: a reduction to an acceptable range of one hundred to two hundred and fifty micrograms per gram, together with a normal CRP.

End of chapter.